Web Data Integration
Project Outline

Pharmaceutical Drug Integration: Linking US, EU and Canadian Approval Data

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October 12, 2026

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1 Use Case

This project aims to build a unified view of approved pharmaceutical drugs in the United States, the European Union and Canada, enabling researchers and analysts to query regulatory status, approval history, and product attributes across these markets from a single integrated data set. To achieve this, we combine data from three complementary sources. Drugs@FDA is the FDA's official database of US-approved drugs. EMA Medicines is the European Medicines Agency's register of EU-authorised medicines. The Health Canada Drug Product Database (DPD) contains current and historical drug products from Health Canada. The central entity is a medicine, represented by its brand name and active substance or combination of active substances. Each integrated record includes the licence holder, regulatory status, product class, ATC code, dosage form, route, strengths, regional regulatory dates and generic, biosimilar or orphan flags.

2 Requirements Suitability

2.1 Schema and basic profile

Table 1 summarises the datasets and their most relevant attributes, including those with high missingness.

Drugs@FDA

Drugs@FDA contains 51,839 product rows across 29,377 applications, forming 8,175 normalized medicine entities. Among applications, 78.3% are ANDAs, 20.1% NDAs, and 1.7% BLAs. Three attributes exceed 30% missingness: `ApplPublicNotes` (99.99%), `SubmissionsPublicNotes` (99.83%), and `ReviewPriority` (41.73%).

EMA Medicines

EMA Medicines contains 2,746 medicine rows, forming 2,620 normalized medicine entities, with no duplicate input rows. Human medicines account for 85.6% of the records. By status, 68.5% are authorised, 15.4% withdrawn, and 10.4% are withdrawn applications. Twelve attributes exceed 30% missingness, mostly because veterinary fields do not apply to human medicines and regulatory-event dates apply only to particular statuses.

Health Canada Drug Product Database (DPD)

The Health Canada DPD contains 58,137 product records across 12 tables linked by `DRUG_CODE`. They are consolidated into 42,042 medicine rows, forming 41,636 normalized medicine entities. Human products account for 87.7% of the records. By status, 66.5% are cancelled and 23.0% marketed. Notable missingness includes ATC code (21.47%) and first marketed date (11.67%).

Table 1: Overview of the input datasets

Dataset	**Source (*)**	**Format**	**# Rows**	**# Entities**	**# Attr.**	**List of attributes (**)**
Drugs@FDA	U.S. FDA	TXT/TSV	51,839	8,175	56	ApplNo, ProductNo, Form, Strength, ReferenceDrug, DrugName, ActiveIngredient, ApplPublic Notes**, SponsorName, Submission-Type, SubmissionStatus, Submission-sPublicNotes**, Review Priority**, MarketingStatusID, TECode, ...
EMA Medicines	EMA	XLSX/CSV	2,746	2,620	39	Name of medicine, EMA product number, Medicine status, Opinion status**, Active substance, Species (vet.)**, ATC code, Biosimilar, Generic, Orphan medicine, Opinion adopted date**, Revision number**, ...
DPD	Health Canada	TXT/CSV	42,042	41,636	87	DIN, brand name, ingredient, strength, dosage unit**, company name, class, ATC code, schedule, status, status date, form, route, descriptor**, ...

(*) Source: Downloaded directly from the respective official provider websites.

Entities: Distinct normalized brand-name and active-substance combinations.
(**) Missingness: ** indicates >30% missing values. High missingness is mostly structural (e.g., optional notes, category-specific fields, or particular regulatory events).

2.2 Integrated schema and overlap with input schemata

The integrated schema contains 15 attributes (Table 2); ten occur in at least two sources. `activeSubstances` and `strengths` are list-valued, while the remaining attributes are single-valued. Strengths are represented as numeric value–unit pairs.

Table 2: Attribute Intersection with Integrated Schema

Attribute name	**Attribute type**	**Datasets in which the attribute is found**
`brandName`	string	Drugs@FDA, EMA Medicines, DPD
`activeSubstances`	list (string)	Drugs@FDA, EMA Medicines, DPD
`licenceHolder`	string	Drugs@FDA, EMA Medicines, DPD
`regulatoryStatus`	string (categorical)	Drugs@FDA, EMA Medicines, DPD
`productClass`	string (categorical)	Drugs@FDA (derived), EMA Medicines, DPD
`atcCode`	string	EMA Medicines, DPD
`generic`	boolean	Drugs@FDA (derived), EMA Medicines
`biosimilar`	boolean	EMA Medicines
`orphanMedicine`	boolean	EMA Medicines
`dosageForm`	string	Drugs@FDA, DPD
`route`	string	Drugs@FDA, DPD
`strengths`	list (numeric value, unit)	Drugs@FDA, DPD
`usApprovalDate`	date	Drugs@FDA
`euAuthorisationDate`	date	EMA Medicines
`caStatusDate`	date	DPD

For consistent counting across the three sources, `medicine_id` combines the normalized `brandName` with the normalized `activeSubstances` list. Text is converted to upper case, accents and dosage-strength expressions are removed, and punctuation and whitespace are standardized. Multiple active substances are deduplicated and sorted before the identifier is formed.

2.3 Entity Count Requirement

Table 1 shows that the entity-count requirement is satisfied. Let F, E and D denote the `medicine_id` values in Drugs@FDA, EMA Medicines and the DPD. The pairwise overlaps are $|F \cap E| = 428$, $|F \cap D| = 949$ and $|E \cap D| = 594$. Their deduplicated union contains $|(F \cap E) \cup (F \cap D) \cup (E \cap D)| = 1{,}407$ distinct medicines, exceeding the required minimum of 1,000.